

Load Interconnection Request System Impact Study Report

For Queue Project Q22-2231

Clayton 138 kV
October 24, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2231. The Interconnection Customer proposes a 121.7 MW large load (data center) facility interconnecting at the Clayton 138 kV substation in the Pennsylvania area. The request was received on November 16, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 471 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,272,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Aford to Empire 138kV
Contingency: Milwaukee to Southbase 138kV Line
Loading: 141.3% (Rating: 327 MVA)
DFAX: 14.3%
Pre-project loading: 136.0%
2. Facility: Asheboro to Empire 138kV
Contingency: Southbase to Milwaukee 138kV
Loading: 120.4% (Rating: 387 MVA)
DFAX: 14.3%
Pre-project loading: 115.9%
3. Facility: North Yuma to Asheboro Substation 138kV
Contingency: William F atson Generating Station to North Yuma 345kV
Loading: 114.4% (Rating: 414 MVA)
DFAX: 5.9%
Pre-project loading: 112.7%
4. Facility: N Yuma 345 kV/138 kV Transformer
Contingency: North Yuma Substation to William F Matson Generating Sta 345kV line
Loading: 112.7% (Rating: 420 MVA)
DFAX: 5.9%
Pre-project loading: 111.0%
5. Facility: Clayton to Greenville Steam Substation 138kV
Contingency: Kings to Clayton 138kV
Loading: 107.7% (Rating: 169 MVA)
DFAX: 45.8%
Pre-project loading: 74.8%

6. Facility: N Yuma to North Allegheny Windpower Project Substation 345kV
 Contingency: Sunrise Center to Merret 500kV
 Loading: 102.5% (Rating: 1,219 MVA)
 DFAX: 11.0%
 Pre-project loading: 101.4%
7. Facility: Southbase to Milwaukee 138kV
 Contingency: Asheboro Substation to Empire 138kV Line
 Loading: 101.1% (Rating: 483 MVA)
 DFAX: 22.4%
 Pre-project loading: 95.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Empire to Axfrd 138kV	Rebuild line with higher-capacity conductor	\$4,264,000	100% (\$4,264,000)
Asheboro to Empire 138kV	Rebuild line with higher-capacity conductor	\$14,036,000	100% (\$14,036,000)
N Yuma to Asheboro 138kV	Rebuild line with higher-capacity conductor	\$5,067,000	100% (\$5,067,000)
N Yuma 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$16,369,000	100% (\$16,369,000)
Clayton to Greenville Steam Substation 138kV	Reconductor line	\$6,019,000	100% (\$6,019,000)
N Yuma to North Allegheny Windpower Project Substation 345kV	Reconductor line	\$20,752,000	29% (\$6,031,000)
Southbase to Milwaukee 138kV	Reconductor line	\$4,486,000	100% (\$4,486,000)
	Total allocated to project		\$56,272,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.